KESHAV RAO

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Full-Stack AI Engineer · LLM Systems & Orchestration · RAG · MCP · React · Python · Cloud

EDUCATION

University of Southern California — Los Angeles, California
Master of Science in Computer Science — December 2025

- *Relevant coursework: Artificial Intelligence, Natural Language Processing, Algorithms, Machine Learning, Distributed Systems*

SRM Institute of Science and Technology — Chennai, India
Bachelor of Technology in Computer Science and Engineering — May 2023

EXPERIENCE

Software Engineer Intern — Dataflix | *Hayward, CA* — August 2025-December 2025

- Shipped **React/TypeScript UI** and **Node.js** REST APIs powering **LLM** customer support agents to production for enterprise clients, serving **~1K** users via IBM watsonx
- Improved response quality by ~**30%** across 3 model families (Granite, Llama-2, Mistral) through structured prompt design, temperature tuning, and beam-search decoding optimization
- Built Node.js services to manage **agentic** orchestration pipelines across multi-turn conversations for concurrent enterprise tenants, cutting context loss errors
- Optimized end-to-end inference latency by ~**15%** through model selection tuning and response caching, enabling real-time agent responses at production scale

Software Research Intern — École de technologie supérieure (ÉTS) | *Montreal, Canada* — July 2022-September 2022

- Designed and implemented **scheduling** algorithms for compute tasks across edge and cloud infrastructure, improving **distributed system** throughput and scalability
- Cut scheduling latency by **20%** and improved simulation pipeline efficiency by **25%** via Monte Carlo Tree Search and Iterative Deepening DFS algorithms
- Scaled the system to **50+** distributed nodes by integrating union-find structures and Van Emde Boas trees within VAPNIC

Research Assistant — SRM Institute of Science and Technology | *Chennai, India* — September 2021-February 2022

- Built ML models for early-stage coronary artery disease prediction, reducing required diagnostic tests.
- Reached **97% classification accuracy** and cut required tests by 50% across 1,000+ patient records using XGBoost, Random Forest & Decision Tree

PROJECTS

AI Knowledge Platform (RAG Microservices + MCP) | *Python, Vector search, React, Elasticsearch, Claude, MCP, OpenAI*

- Built a full-stack **agentic** knowledge assistant, **React** frontend, RESTful FastAPI backend for natural language queries over internal docs via **MCP** and Elasticsearch
- Developed TypeScript/Node.js middleware using **Elasticsearch** as a dual-purpose **vector** store and keyword search engine, with embedding, chunking, and reranking via OpenAI and Claude over MCP, achieving <1.5s **retrieval latency**
- Reduced **hallucinations** through retrieval evaluation, iteratively tuning chunk size, overlap, and prompt templates

Recruiter Automation Platform | *Python, PyTorch, React, TypeScript, Node.js*

- Built an end-to-end generative pipeline (React/Node.js + GPU-accelerated PyTorch + Stable Diffusion XL) that produces personalized recruiter reach out mails and marketing assets at scale, cutting asset creation time by **20%** via persistent model caching and batch **optimization**

Phishing Webpage Detection System | *TensorFlow, Keras, React, TypeScript*

- Built a real-time phishing detection model (TensorFlow/Keras) achieving **92%** accuracy (+3.5% over baseline) via hyperparameter tuning and scheduling, with a React/TypeScript dashboard via REST API for live prediction visualization

PUBLICATIONS

B. Balakrishnan, K. Kumaran, A. Sankar and **K. Rao**, "Coronary Artery Disease Prediction using Machine Learning Algorithms," IEEE ICESIC 2022, pp. 95-100. doi: 10.1109/ICESIC53714.2022.9783592

SKILLS

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, C
Frameworks: React, Node.js, FastAPI, TensorFlow, PyTorch, Hugging Face
AI Systems: LLMs, RAG Systems, Generative AI, Inference Pipelines, semantic search, Elasticsearch
Cloud: AWS, GCP, IBM watsonx, Docker, Git